

Self-Supervised Anomaly Detection With Neural Transformations

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Abstract—Data augmentation plays a critical role in self-supervised learning, including anomaly detection. While hand-crafted transformations such as image rotations can achieve impressive performance on image data, effective transformations of non-image data are lacking. In this work, we study *learning* such transformations for end-to-end anomaly detection on arbitrary data. We find that a contrastive loss—which encourages learning diverse data transformations while preserving the relevant semantic content of the data—is more suitable than previously proposed losses for transformation learning, a fact that we prove theoretically and empirically. We demonstrate that anomaly detection using neural transformation learning can achieve state-of-the-art results for time series data, tabular data, text data and graph data. Furthermore, our approach can make image anomaly detection more interpretable by learning transformations at different levels of abstraction.

Index Terms—Deep anomaly detection, self-supervised anomaly detection, self-supervised learning, neural transformations.

I. INTRODUCTION

MANY recent advances in anomaly detection rely on the paradigm of data augmentation. In the self-supervised setting, especially for image data, predefined transformations such as rotations, reflections, and cropping are used to generate varying *views* of the data. This idea has led to strong anomaly detectors based on either transformation prediction [25], [31],

[82] or using representations learned using these views [15] for downstream anomaly detection tasks [74], [75].

Unfortunately, for data other than images, such as time series, tabular data, graphs, or text, it is much less well known which transformations are useful, and it is hard to design these transformations manually. This paper studies self-supervised anomaly detection for general data types. We develop Neural Transformation Learning for Anomaly Detection (NeuTraL AD): a simple end-to-end procedure for anomaly detection with *learnable* transformations. Instead of manually designing data transformations to construct auxiliary prediction tasks that can be used for anomaly detection, we derive a single objective function for jointly learning useful data transformations and anomaly scoring. As detailed below, the idea is to learn a variety of transformations such that the transformed samples share semantic information with their original form while different views are easily distinguishable.

NeuTraL AD has only two components: a fixed set of learnable transformations and an encoder model. Both elements are jointly trained on a noise-free Deterministic Contrastive Loss (DCL) designed to learn faithful transformations. Our DCL is different from other contrastive losses in representation learning [8], [15], [28], [50], [56] and image anomaly detection [74], [75], all of which use negative samples from a noise distribution. In contrast, our approach constructs other learnable transformations as negative samples and leads to a non-stochastic objective that neither needs any additional regularization nor adversarial training [76] and can be directly used as the anomaly score.

In Section II we describe NeuTraL AD and DCL, a novel objective for learning diverse and semantically meaningful transformations. We also establish the connections between NeuTraL AD and popular alternative deep anomaly detection method, specifically deep One-Class Classification (OCC). We show that NeuTraL AD generalizes deep OCC and has several advantages thanks to the learnable transformations.

Our approach leads to a new state-of-the-art in deep anomaly detection. For time series and tabular data, NeuTraL AD significantly improves the anomaly detection accuracy. For example, on an epilepsy time series dataset, we raised the state-of-the-art from an AUC of **82.6%** to **92.6%** (+10%). On an arrhythmia tabular dataset, we raised the F1-score by 3.7 percentage points. In the case of anomaly detection on text, we raised the state-of-the-art (no outlier exposure) from an AUC of 93.4% to 94.7% (+1.3%) on the Reuters dataset. For graph-level anomaly detection, we raised the bar in terms of AUC by 7.5 percentage points averaged over six datasets from various domains. In terms

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The code is available at <https://github.com/boschresearch/NeuTraL-AD.git>.

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of images, NeuTraL AD generalizes self-supervised anomaly detection to image features and achieves comparable results to the state-of-the-art method.

We begin by presenting NeuTraL AD in Section II. We then demonstrate the effectiveness of our method with comprehensive experiments in Section III. Finally, we discuss related work in Section IV and conclude the paper in Section V.

Our previous work, Qiu et al. [60], introduced NeuTraL AD for deep anomaly detection beyond images, focusing primarily on time series and tabular data. In that study, we proposed using a fixed set of learnable transformations and an encoder model, jointly trained with a noise-free contrastive loss. The approach was limited in its application to specific data types and did not fully explore the potential of the model formulation.

In this paper, we extend NeuTraL AD in several key ways:

- *Generalized Model Formulation:* We broaden the model by exploring various similarity functions in the loss, showing that deep one-class classification is a special case of our framework.
- *Extension to New Data Modalities:* We expand NeuTraL AD to handle images, text, and graphs, demonstrating its versatility and effectiveness across different domains.
- *Comprehensive Theoretical and Empirical Analysis:* We provide a detailed analysis of the DCL, including theoretical justification and empirical studies that balance the objectives of aligning with original data embeddings while encouraging diversity among transformed views.
- *Extensive Experiments and Ablation Studies:* We conduct comprehensive experiments and ablation studies that offer deeper insights into the model's effectiveness and robustness across various datasets and data types including time series, tabular data, images, text, and graphs.

These advancements represent significant contributions beyond our earlier work, offering a more comprehensive understanding and broader applicability of NeuTraL AD.

II. ALGORITHM AND METHODOLOGY

We develop neural transformation learning for anomaly detection (NeuTraL AD), a deep anomaly detection method based on contrastive learning for general data types. It is a simple pipeline with two components: a set of learnable transformations and a loss function. The transformations are trained jointly on a DCL. The objective has two purposes. During training, it is optimized to find the parameters of the transformations. During testing, it is also used to score each sample as either an inlier or an anomaly. We now describe our approach in detail.

A. Preliminaries

The essence of an anomaly detection algorithm is to produce a score function $s(x) : \mathcal{X} \rightarrow \mathbb{R}$, which measures the degree of abnormality of each sample in the data space $\mathcal{X}$. The score function is used during test time to detect anomalies by evaluating whether the score is above a threshold τ_a ,

$$\hat{y} = \begin{cases} 1, & \text{if } s(x) \geq \tau_a \quad \text{meaning } x \text{ abnormal,} \\ 0, & \text{otherwise} \quad \text{meaning } x \text{ normal.} \end{cases} \quad (1)$$

TABLE I
MAIN NOTATIONS USED IN THE ALGORITHM

<u>Indices:</u>	
K	number of transformations ($k \in \{1, \dots, K\}$)
N	number of data samples ($i \in \{1, \dots, N\}$)
<u>Variables:</u>	
x	data sample
$\hat{y}$	predicted label of being an anomaly
$\mathcal{X}$	data space
$\mathcal{Z}$	embedding space
$\mathcal{L}$	deterministic contrastive loss (DCL)
<u>Functions:</u>	
$T_k(\cdot)$	neural transformation function $\mathcal{X} \rightarrow \mathcal{Z}$
$h(\cdot, \cdot)$	similarity function between two variables $\mathcal{Z} \rightarrow \mathbb{R}$
$l(\cdot)$	loss function $\mathcal{X} \rightarrow \mathbb{R}$
$s(\cdot)$	anomaly score function $\mathcal{X} \rightarrow \mathbb{R}$
<u>Parameters:</u>	
τ_a	estimated threshold for detecting anomalies
τ	temperature controlling the sensitivity of the similarity measure
θ_k	learnable parameters of the transformation T_k

The threshold τ_a is calibrated for the specific application (e.g., using a labeled validation set). A data-driven anomaly detection algorithm relies on an unlabeled training dataset $\mathcal{D} = \{x_1, \dots, x_N\} \subset \mathcal{X}$ to learn a score function. Here we consider a self-supervised approach.

Self-supervised methods typically learn a feature extractor and representations by solving auxiliary tasks. The tasks are typically formulated by defining a training loss. In self-supervised anomaly detection [25], task performance (i.e., the loss function used during training) is directly used as the score function. For example, in Golan and El-Yaniv [25] and Wang et al. [82] image transformations such as rotations, blurring, and cropping are applied to input images. The learning task consists of predicting which transformation has been applied given a transformed input image. This task is formalized in a loss function (e.g., cross-entropy of prediction). During training, the loss will decrease on most training examples. Due to the principle of *inlier priority* [82] and assuming that most training examples are normal inliers, training will promote the model to generalize better to held-out inliers than anomalies. The loss on a held-out example can therefore be used as the anomaly score.

Our approach also follows the principles of self-supervised anomaly detection. We next develop the NeuTraL AD training task which defines our approach.

B. Methodology

NeuTraL AD learns an anomaly detector which contains $K + 1$ transformations $\mathcal{T} := \{T_0, T_1, \dots, T_K\}$. Each transformation is a parameterized function (e.g., a neural network) $T_k(\cdot; \theta_k) : \mathcal{X} \rightarrow \mathcal{Z}$ that maps samples into an embedding space $\mathcal{Z}$. We sometimes drop the parameter dependence for brevity, $T_k(\cdot; \theta_k) \equiv T_k(\cdot)$. These embeddings include one reference embedding $T_0(x)$ and K transformed embeddings $T_k(x)$ for $k = 1, \dots, K$. We assume that all transformations are learnable, i.e., they can be modeled by any parameterized function whose parameters θ_k are accessible to gradient-based optimization. A schematic of NeuTraL AD is in Fig. 1. Each sample is transformed by a set of learnable transformations and then embedded

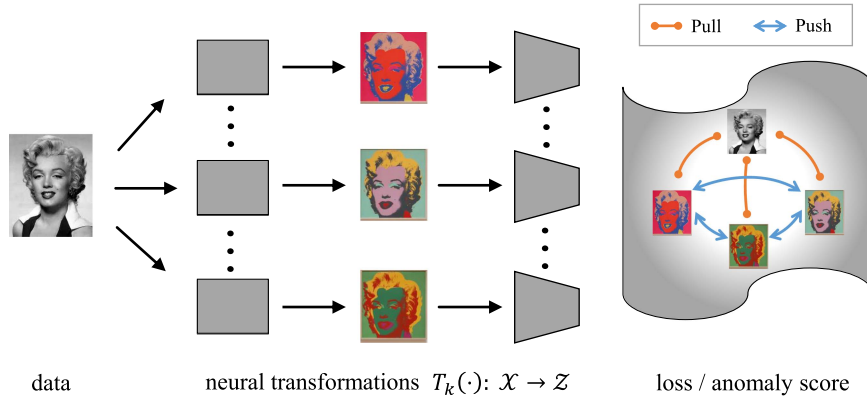


Fig. 1. NeuTraL AD is an end-to-end procedure for self-supervised anomaly detection with *learnable* neural transformations $T_k(\cdot) : \mathcal{X} \rightarrow \mathcal{Z}$. Each sample is transformed by a set of learned transformations and then embedded into a semantic space. The neural transformations are trained jointly on a contrastive objective (3), which is also used to score anomalies.

into a semantic space such that (1) all transformed embeddings preserve some semantic information about the input sample, while (2) all transformations remain distinct from another. As follows, we motivate NeuTraL AD’s loss function and its usage in anomaly scoring.

1) *Loss Function*: A key ingredient of NeuTraL AD is a new loss function that we call “Deterministic Contrastive Loss” (DCL), explained below. This loss is constructed such as to encourage the transformations to learn salient features of the training data (which is assumed to belong to the normal class). The DCL encourages each transformed embedding to be similar to the reference embedding while encouraging it to be dissimilar from other transformed embeddings of the same sample. We define the similarity function of two embeddings $a, b \in \mathcal{Z}$ as

$$h(a, b) = \exp(\text{sim}(a, b)/\tau), \quad \text{sim}(a, b) := a^T b / \|a\| \|b\|, \quad (2)$$

where τ denotes a temperature parameter. Unless mentioned otherwise, we work with cosine similarity as $\text{sim}(a, b)$. By plugging $K + 1$ transformations and averaging over all data samples, DCL is defined using the similarity function h as

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N l(x_i) \quad \text{with} \quad l(x) := - \sum_{k=1}^K \log \frac{h(T_k(x), T_0(x))}{\sum_{T_l \in \mathcal{T} \setminus \{T_k\}} h(T_k(x), T_l(x))}. \quad (3)$$

Intuitively, the term in the nominator pulls the transformed embeddings close to the reference embedding. This encourages the transformed embeddings to preserve relevant semantic information of the reference embedding. The denominator pushes all transformed embeddings away from each other, thereby encouraging diverse transformations. We empirically demonstrate that the DCL manages to balance these two competing objectives in an ablation in Section III-B4. A detailed explanation on the trade-off between the loss terms can be found in Appendix A.4, available online.

2) *Optimization*: The parameters of all transformations $\theta_{0,\dots,K}$ are optimized jointly with stochastic gradient descent or its variants (e.g., Adam [38]). The optimal parameters $\theta_{0,\dots,K}^*$

are computed by minimizing (3) on the training samples as

$$\theta_{0,\dots,K}^* = \underset{\theta_{0,\dots,K}}{\text{argmin}} - \sum_{i=1}^N \sum_{k=1}^K \log \frac{h(T_k(x_i), T_0(x_i))}{\sum_{T_l \in \mathcal{T} \setminus \{T_k\}} h(T_k(x_i), T_l(x_i))}. \quad (4)$$

This training objective leads to neural transformations that produce diverse views of each sample. Next, we describe how it is used to detect anomalies.

3) *Score Function*: One advantage of our approach over other methods is that our loss function is also our anomaly score. We define an anomaly score $s(x)$ as

$$s(x) := l(x). \quad (5)$$

By minimizing the DCL (3), we minimize the score for training examples (inliers). The transformations learn to highlight salient features of the data such that a low loss can be achieved. After training, samples from the normal class have a low anomaly score, while anomalies are handled less well by the model and thus have a high score. The higher the anomaly score, the more likely that a sample is an anomaly. Unlike most other contrastive losses in representation learning [15], [28] and image anomaly detection [74], [75], the “negative samples” are not drawn from a noise distribution (e.g., other samples in the minibatch) but constructed deterministically from x in our loss and score function. Dependence on the minibatch for negative samples would need to be accounted for at test time. Drawing negative samples from the test data is biased, and using the training data is infeasible in practice. In contrast, the deterministic nature of (5) makes it a simple choice for anomaly detection.

This concludes the proposed method NeuTraL AD, an end-to-end procedure for transformation learning and anomaly detection. The detailed training process and test process of NeuTraL AD are in Algorithm 1. We stress that it is simple and effective without the need of any additional regularization.

Qiu et al. [60] provide several theoretical arguments why NeuTraL AD is better suited for transformation learning than alternative loss functions. The theorems in Qiu et al. [60] which are included with proofs in Appendix A, available online state

Algorithm 1: Algorithm of NeuTraL AD.

Data: training dataset $\mathcal{D}$, test data x_t
Model: neural transformations
 $\mathcal{T} := \{T_0, \dots, T_K \mid T_k(\cdot; \theta_k)\}$
// Training process
foreach *epoch* **do**
 Transform each training sample
 $T_0(x_i), \dots, T_K(x_i) \leftarrow x_i \in \mathcal{D}$
 Minimize the loss $\mathcal{L}$ in Equation (3) to update
 $\theta_{0, \dots, K}$
end
// Test process
 Transform the test data $T_0(x_t), \dots, T_K(x_t) \leftarrow x_t$
 Calculate the anomaly score as in Equation (5)

that optimizing certain existing losses for self-supervised learning w.r.t. the transformations leads to trivial edge-cases, while these edge-cases are not optimal under the DCL objective. It follows, that the DCL does not produce these trivial edge cases and is suited for transformation learning and anomaly detection. Next, we describe how it relates to one-class classification.

C. Connection to One-Class Classification

One of the most influential deep anomaly detection methods is deep OCC due to its simplicity and applicability to various types of data [26], [44], [64]. Below we explore how NeuTraL AD is connected to deep OCC. We show that NeuTraL AD can be interpreted as a generalization of deep OCC. We also discuss why NeuTraL AD is more powerful and more flexible than deep OCC.

The main idea of deep OCC [64] is to use a parametrized function (e.g., neural network) $T(\cdot) : \mathcal{X} \rightarrow \mathcal{Z}$ to map input data into a latent space $\mathcal{Z}$. Every input datum x_i is mapped as close as possible to the same point $c \in \mathcal{Z}$ termed “the center”. The loss function, also used for scoring anomalies, is

$$\mathcal{L}_{\text{OCC}} := \frac{1}{N} \sum_{i=1}^N \|T(x_i) - c\|_2^2. \quad (6)$$

The center has to be fixed in advance to avoid a trivial solution [64] and is typically set to the mean of the embeddings obtained by an initial forward pass.

We can specialize DCL so that it has a similar form to $\mathcal{L}_{\text{OCC}}$. To this end, we set the similarity function to the negative squared euclidean distance, $\text{sim}(a, b) := -\|a - b\|_2^2$. This results in a multi-view generalization of deep OCC involving multiple mappings $T_k(\cdot) : \mathcal{X} \rightarrow \mathcal{Z}$,

$$\begin{aligned} \mathcal{L}_{\text{MOCC}} &:= -\frac{\tau}{N} \sum_{i=1}^N \sum_{k=1}^K \log \frac{\exp(-\|T_k(x_i) - T_0(x_i)\|_2^2/\tau)}{Z_{k,i}} \\ &= \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \|T_k(x_i) - T_0(x_i)\|_2^2 + \tau \log Z_{k,i}, \end{aligned}$$

$$\text{where } Z_{k,i} = \sum_{T_l \in \mathcal{T} \setminus \{T_k\}} \exp(-\|T_k(x_i) - T_l(x_i)\|_2^2/\tau). \quad (7)$$

After rearranging the loss terms, its connection to deep OCC becomes apparent. $\mathcal{L}_{\text{MOCC}}$ ties together K OCC-type losses and a regularization term. Instead of a fixed center c , the one-class terms now have datapoint-dependent centers $T_0(x_i)$. There are multiple embeddings $T_k(x_i)$ of each data point, and the regularizer $\log Z_{k,i}$ prevents all embeddings from being the same. For $K = 1$ and removing $\log Z_{k,i}$ (or sending $\tau \rightarrow 0$), $\mathcal{L}_{\text{MOCC}}$ is equivalent to $\mathcal{L}_{\text{OCC}}$ when $T_0(x)$ is replaced by a constant, independent of x_i .

Discussion. We empirically find that $\mathcal{L}_{\text{MOCC}}$ is more powerful than $\mathcal{L}_{\text{OCC}}$ (see Section III), which could be explained as follows. First, compared to deep OCC, NeuTraL AD learns to extract multiple, different features for anomaly detection (as opposed to only one feature) by including multiple learnable views. All extracted features are enforced to be diverse by the regularization term. Second, $T_0(x)$ serves as a datapoint-dependent center. Compared with the fixed center c in $\mathcal{L}_{\text{OCC}}$, $T_0(x)$ is more flexible since it can be optimized jointly with the other parameters. $T_0(x)$ is also more informative since it preserves instance-level information beyond the common factors of variation in the dataset at large. Interestingly, the DCL does not couple any data points explicitly, while $\mathcal{L}_{\text{OCC}}$ couples the data points by pulling them to a shared center c . NeuTraL AD seeks to find “intrinsic” features within learnable views that characterize the normality of data. By measuring the similarity of each view/feature with the reference embedding $T_0(x)$, NeuTraL AD detects the anomalies.

Next, we present an empirical study on five application domains to show the benefit of NeuTraL AD over state-of-the-art (deep) anomaly detection methods.

III. APPLICATIONS

NeuTraL AD is applicable to many application domains: it can be used to detect abnormal time series, it is applicable to anomaly detection on tabular data, and it can be applied to graphs. The advantages of NeuTraL AD include that it outperforms other deep anomaly detection methods (as our experimental results in this section will show) and that it does not require hand-crafted data augmentation schemes for specialized domains. Since the neural transformations are *learned* together with the other architecture components, they automatically lead to an appropriate data augmentation scheme. In fact, since the transformations do not need to be hand-crafted, they can even be applied to data representations that are not easily accessible to human intuition, such as image features that are extracted from an image using a pretrained neural network, or to word embeddings from a language model. This allows us to use NeuTraL AD to also perform anomaly detection on images and on text.

In this section, we present the following applications of NeuTraL AD:

- 1) Section III-B: Time-series anomaly detection. We evaluate NeuTraL AD on identifying whole abnormal sequences. We find that it can effectively find anomalies in audio signals, health care signals, as well as motion signals.
- 2) Section III-C: Anomaly detection on visual data. We study NeuTraL AD on raw images and image features.

- 3) Section III-D: Anomaly detection on tabular data. We find that NeuTraL AD is the most effective method for detecting anomalies in four different datasets from the medical and cyber security domains.
- 4) Section III-E: Anomaly detection on text. We build NeuTraL AD on word embeddings to detect whole abnormal sentences in the Reuters dataset.
- 5) Section III-F: Graph-level anomaly detection. We assess how well NeuTraL AD detects whole abnormal graphs in six datasets from bioinformatics, molecular, and social-networks domains.

For each application, we specify the choice of the neural transformations $T_k(\cdot)$ and the similarity function $\text{sim}(a, b)$ included in the DCL. We present an extensive empirical evaluation in comparison with other successful approaches for anomaly detection and discuss the results and their implications.

A. Evaluation Protocol

An anomaly detection method is useful if it is good at detecting anomalies in unseen test data. To test this, we follow the standard evaluation protocol for anomaly detection in the literature e.g., [1], [4], [9], [18], [25], [31], [37], [58], [64], [65], [74], [75], [80]. Since in many domains it is hard to find labeled test sets for anomaly detection, a common approach for setting up an evaluation pipeline is to repurpose classification datasets. We consider two such evaluation protocols: the standard ‘one-vs-rest’ and the more challenging ‘ n -vs-rest’ evaluation protocol. Both settings turn a classification dataset into a quantifiable anomaly detection benchmark.

a) One-vs-Rest: This evaluation setup has been used in virtually all recent papers on deep anomaly detection published at top-tier venues e.g., [1], [4], [9], [18], [25], [31], [37], [58], [64], [65], [74], [75], [80]. For ‘one-vs.-rest’, the dataset is split by the $\mathcal{N}$ class labels, creating $\mathcal{N}$ one-class classification task, in each of which one class is considered normal, and only samples drawn from this class are used as training data. All other classes are considered abnormal. The test and validation sets contain samples from all classes including the normal class. The samples from the other classes should be detected as anomalies. By iterating over the classes and changing which class is considered normal, we obtain $\mathcal{N}$ separate anomaly detection tasks for evaluation.

b) N -vs-rest: An alternative evaluation protocol we also consider follows the insight that it is more realistic in practice that the “normal distribution” is more diverse than a single class (i.e., it is more realistic that it contains multiple classes, Deecke et al. [2], [19]). So we also evaluate methods on the more challenging n -vs.-rest protocol, where n classes (for $1 < n < \mathcal{N}$) are treated as normal, and the remaining classes provide the anomalies in the test and validation set. By increasing the variability of what is considered normal data, anomaly detection becomes more challenging.

B. Anomaly Detection on Time Series

Our goal is to detect abnormal time series on a *whole-sequence level*. This is a different set-up than anomaly detection *within*

time series but equally important in practice. For example, one might want to detect abnormal sound or find production quality issues by detecting abnormal sensor measurements recorded over the duration of producing a batch. Other applications include sports and health monitoring; e.g., finding abnormal movement patterns during sports can be indicative of fatigue or injury.

We first present the datasets and baselines used to study NeuTraL AD. We then present the implementation details and finally describe the empirical results. We find that NeuTraL AD learns meaningful and diverse transformations and detects anomalous time series successfully.

1) Datasets and Baselines. Time Series Datasets: We select datasets from various domains. The datasets come from the UEA multivariate time series classification archive¹ [7]. We evaluate NeuTraL AD on them with both one-vs-rest setting and n -vs-rest setting.

- *Spoken Arabic Digits (SAD):* Sound of ten Arabic digits, spoken by 88 speakers. The samples are stored as 13 Mel Frequency Cepstral Coefficients. We select sequences with the length between 20 and 50 and get a dataset of 7824 samples. The sequences that are shorter than 50 are zero padded to have the length of 50.
- *Naval Air Training and Operating Procedures Standardization (NATOPS):* The data is from a motion detection competition of various movement patterns used to control planes in naval air training. The data has six classes of distinct actions. The dataset has 360 samples, each being a sequence of x, y, z coordinates for eight body parts of length 51.
- *Character Trajectories (CT):* The data consists of 2858 character samples from 20 classes. Each instance is a 3-dimensional pen tip velocity trajectory. The data is truncated to the length of the shortest, which is 182.
- *Epilepsy (EPSY):* The data was generated with healthy participants simulating four different activities: walking, running, sawing with a saw, and seizure mimicking whilst seated. The dataset has 275 samples, each being a 3-dimensional sequence of length 203.
- *Racket Sports (RS):* The data is a record of university students playing badminton or squash. The data records the x, y, z coordinates for both the gyroscope and accelerometer. Sport and stroke types separate the data into four classes. The dataset has 303 samples, each being a 6-dimensional sequence with a length of 30.

Time Series Baselines: We study NeuTraL AD in comparison to unsupervised and self-supervised anomaly detection methods. They include three anomaly detection baselines: The One-Class Support Vector Machine (OCSVM) [48], Isolation Forest (IF) [43], a tree-based model which aims to isolate anomalies, and Local Outlier Factor (LOF) [11], which uses density estimation with K -Nearest Neighbor (KNN).

¹We selected datasets on which supervised multi-class classification methods achieve strong results [68]. Only datasets with separable classes can be repurposed for anomaly detection.

Next, we include two deep anomaly detection methods, Deep Support Vector Data Description (deep SVDD) [64], which fits a one-class classification in the feature space of a neural net, and Deep Autoencoding Gaussian Mixture Model (DAGMM) [93], which estimates the density in the latent space of an autoencoder. We also include two baselines specifically designed for time series data: The RNN-Based Model (RNN) directly models the data distribution $p(x_{1:T}) = \prod p(x_t|x_{<t})$ and uses the log-likelihood as the anomaly score. LSTM-Based Encoder-Decoder (LSTM-ED) [47] is an encoder-decoder time series model where the anomaly score is based on the reconstruction error.

Finally, We choose two self-supervised baselines, which are technically also deep anomaly detection methods, deep support vector data description (deep SVDD) [9] is a distance-based classification method based on random affine transformations. Golan and El-Yaniv [25] is a softmax-based classification method based on hand-crafted transformations, which show impressive performance on images. We adopt their pipeline to time series here by crafting specific time series transformations (fixed Ts). Their implementation details are provided in Appendix A, available online.

2) *Neural Transformation Learning on Time Series*: We consider a dataset of time series $x \in \mathbb{R}^{d \times l}$, where d is the data dimension and l is the time series length. T_0 is modeled by a composition of an identity transformation and an encoder f . The transformations T_k with $k \in \{1, \dots, K\}$ are modeled by composing transformation functions ϕ_k with the shared encoder f . We consider two parametrizations of the transformation function ϕ_k : *residual* $\phi_k(x) := M_k(x) + x$ with $M_k(x) \in \mathbb{R}^d$, and *multiplicative* $\phi_k(x) := M_k(x) \odot x$ with $M_k(x) \in (0, 1)^d$. Both M_k and the encoder f are modeled by convolutional neural networks. In the embedding space, we define the similarity function as the cosine similarity $\text{sim}(a, b) := a^T b / \|a\| \|b\|$.

Implementation Details: M_k is a neural network that consists of one 1d convolutional layer on the bottom, a stack of three residual blocks of 1d convolutional layers with affine-free instance normalization layers and ReLU activations, as well as one 1d convolutional layer on the top. All bias terms in the network are fixed as zero. The network of the encoder f consists of several residual blocks of 1d convolutional layers with a hidden dimension of 32, as well as one final linear layer to extract the embeddings. The number of residual blocks depends on the data dimension. Specifically, we use four residual blocks in the encoder for RS, five residual blocks for SAD and NATOPS, seven residual blocks for CT and Epilepsy (EPSY). The output dimensions of the encoders are 32 for SAD, 128 for EPSY, and 64 for others. On all time series datasets, we set the number of transformations $K = 11$.

3) *Empirical Results*: The results of NeuTraL AD in comparison to the baselines on time series datasets from various fields are reported in Table II. NeuTraL AD raises the detection accuracy in terms of AUC by 7.2% on average. NeuTraL AD outperforms all shallow baselines in all experiments and outperforms the deep learning baselines in 4 out of 5 experiments. Only on the RS data, it is outperformed by transformation prediction with fixed transformations, which we designed to understand

TABLE II
AVERAGE AUCs (%) WITH STANDARD DEVIATION FOR ONE-VS-REST ANOMALY DETECTION ON TIME SERIES DATASETS

	SAD	NATOPS	CT	EPSY	RS	Avg.
OCSVM	95.3	86.0	97.4	61.1	70.0	82.0
IF	88.2	85.4	94.3	67.7	69.3	81.0
LOF	98.3	89.2	97.8	56.1	57.4	79.8
RNN	81.5 $\pm$ 0.4	89.5 $\pm$ 0.4	96.3 $\pm$ 0.2	80.4 $\pm$ 1.8	84.7 $\pm$ 0.7	86.5
LSTM-ED	93.1 $\pm$ 0.5	91.5 $\pm$ 0.3	79.0 $\pm$ 1.1	82.6 $\pm$ 1.7	65.4 $\pm$ 2.1	82.3
deep SVDD	86.0 $\pm$ 0.1	88.6 $\pm$ 0.8	95.7 $\pm$ 0.5	57.6 $\pm$ 0.7	77.4 $\pm$ 0.7	81.1
DAGMM	80.9 $\pm$ 1.2	78.9 $\pm$ 3.2	89.8 $\pm$ 0.7	72.2 $\pm$ 1.6	51.0 $\pm$ 4.2	74.6
GOAD	94.7 $\pm$ 0.1	87.1 $\pm$ 1.1	97.7 $\pm$ 0.1	76.7 $\pm$ 0.4	79.9 $\pm$ 0.6	87.2
fixed Ts	96.7 $\pm$ 0.1	78.4 $\pm$ 0.4	97.9 $\pm$ 0.1	80.4 $\pm$ 2.2	87.7 $\pm$ 0.8	88.2
NeuTraL AD (ours)	98.9 $\pm$ 0.1	94.5 $\pm$ 0.8	99.3 $\pm$ 0.1	92.6 $\pm$ 1.7	86.5 $\pm$ 0.6	94.4

NeuTraL AD achieves the best results on 4 of 5 datasets.

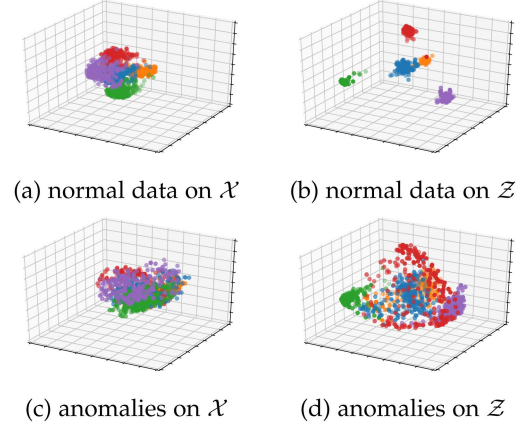


Fig. 2. 3D visualizations (projected using PCA) of different views in the data space and the embedding space of the encoder. The original samples (blue) from the SAD dataset and the different views created by the neural transformations (one color per transformation type) cluster in the data space [Fig. 2(a) and (c)] and in the embedding space [Fig. 2(b) and (d)]. The crisp separation of the different transformations of held-out inliers [Fig. 2(b)] in contrast to the overlap between transformed anomalies [Fig. 2(d)] visualizes how NeuTraL AD is able to detect anomalies.

the value of learning transformations with NeuTraL AD vs using hand-crafted transformations. The results confirm that designing the transformations only succeeds sometimes, whereas with NeuTraL AD we can learn the appropriate transformations. The learned transformations also give NeuTraL AD a competitive advantage over the other self-supervised baseline GOAD which uses random affine transformations. The performance of the traditional anomaly detection baselines hints at the difficulty of each anomaly detection task; the traditional methods perform well on SAD and CT, but perform worse than the deep learning based methods on other data.

What does NeuTraL AD learn? For visualization purposes, we train NeuTraL AD with the learnable transformations on the SAD data. Fig. 2 shows the structure in the data space $\mathcal{X}$ and the embedding space of the encoder $\mathcal{Z}$ after training. Held-out data samples (blue) are transformed by each of the learned transformations with the multiplicative transformation function $\phi_k(x) = M_k(x) \odot x$ to produce $K = 4$ different views of each sample (the transformations are color-coded by the other colors). Projection to three principal components with PCA allows for visualization in 3D. In Fig. 2(a) and (c), we can see that the transformations already cluster together in the data space, but with the help of the encoder, the different views of

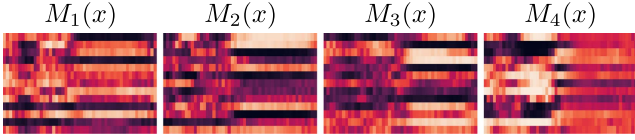


Fig. 3. Heat maps of four learned masks $M_{1:4}$ for SAD spectrograms. The dark horizontal lines indicate where M_1 and M_2 mask out frequency bands almost entirely, while the bright spot in the middle left part of M_4 indicates that this mask brings the intermediate frequencies in the first half of the recording into focus. NeuTraL AD learns dissimilar transformations.

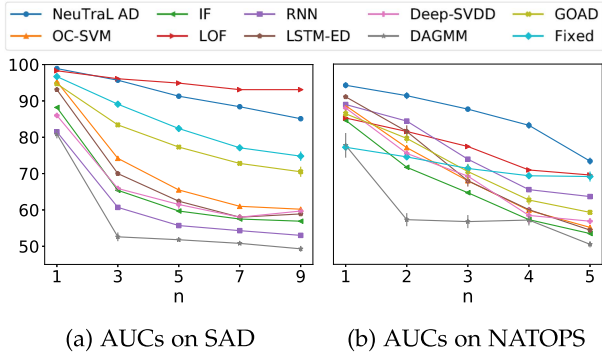


Fig. 4. AUCs of n -vs-all experiments on time series datasets SAD and NATOPS with error bars. NeuTraL AD (ours) outperforms all baselines on NATOPS and all deep learning baselines on SAD. LOF, a method based on k -nearest neighbors, outperforms NeuTraL AD, when $n > 3$ on SAD. NeuTraL AD is more robust to the variability of inliers than deep learning baselines.

inliers are separated from each other (Fig. 2(b)). In comparison, the anomalies and their transformations are less structured in $\mathcal{Z}$ (Fig. 2(d)), visually explaining why they incur a higher anomaly score and can be detected as anomalies.

The learned masks $M_{1:4}(x)$ of one inlier x are visualized in Fig. 3. We can see that the four masks are dissimilar from each other and have learned to focus on different aspects of the spectrogram. The masks take values between 0 and 1, with dark areas corresponding to values close to 0 that are zeroed out by the masks, while light colors correspond to the areas of the spectrogram that are not masked out. Interestingly, in M_1 , M_2 , and M_3 , we can see ‘black lines’ where they mask out entire frequency bands at least for part of the sequence. In contrast, M_4 has a bright spot in the middle left part of the spectrogram; it creates views that focus on the content of the intermediate frequencies in the first half of the recording.

How do the methods cope with an increased variability of inliers? To study this empirically, we increase the number of classes n considered to be inliers. We test all methods on SAD and NATOPS under the n -vs-rest setting with varying n consecutive classes. From Fig. 4 we can observe that the performance of all methods drops as the number of classes included in the normal data increases. This shows that the increased variance in the nominal data makes the task more challenging. NeuTraL AD outperforms all baselines on NATOPS and all deep-learning baselines on SAD. It is interesting that LOF, a method based on KNN, performs better than our method (and all other baselines) on SAD when n is larger than three.

TABLE III
AVERAGE AUCs (%) WITH STANDARD DEVIATION FOR n -VS-REST ($n = \mathcal{N} - 1$) ANOMALY DETECTION ON TIME SERIES DATASETS

	SAD	NATOPS	CT	EPSY	RS	Avg.
OCSVM	60.2	57.6	57.8	50.2	55.9	56.3
IF	56.9	56.0	57.9	55.3	58.4	56.9
LOF	93.1	71.2	90.3	54.7	59.4	73.7
RNN	53.0 $\pm$ 0.1	65.6 $\pm$ 0.4	55.7 $\pm$ 0.8	74.9 $\pm$ 1.5	75.8 $\pm$ 0.9	64.9
LSTM-ED	58.9 $\pm$ 0.5	56.9 $\pm$ 0.7	50.9 $\pm$ 1.2	56.8 $\pm$ 2.1	63.1 $\pm$ 0.6	57.3
deep SVDD	59.7 $\pm$ 0.5	59.2 $\pm$ 0.8	54.4 $\pm$ 0.7	52.9 $\pm$ 1.4	62.2 $\pm$ 2.1	57.7
DAGMM	49.3 $\pm$ 0.8	53.2 $\pm$ 0.8	47.5 $\pm$ 2.5	52.0 $\pm$ 1.0	47.8 $\pm$ 3.5	50.0
GOAD	70.5 $\pm$ 1.4	61.5 $\pm$ 0.7	81.1 $\pm$ 0.1	62.7 $\pm$ 0.9	68.2 $\pm$ 0.9	68.8
fixed Ts	74.8 $\pm$ 1.3	70.8 $\pm$ 1.3	63.0 $\pm$ 0.6	69.8 $\pm$ 1.6	81.6$\pm$1.2	72.0
NeuTraL AD (ours)	85.1 $\pm$ 0.3	74.8$\pm$0.9	87.4 $\pm$ 0.2	80.5$\pm$1.0	80.0 $\pm$ 0.4	81.6

We bold the top two results. NeuTraL AD outperforms deeplearning based baselines.

We include quantitative results for $n = \mathcal{N} - 1$ under the n -vs-rest setting for all time series datasets, where only one class is considered abnormal, and the remaining $\mathcal{N} - 1$ classes are normal. As shown in Table III, NeuTraL AD raises the detection AUC by 7.9% on average and outperforms other deep learning methods on 4 out of 5 datasets. On RS, it is just outperformed by transformation prediction with hand-crafted transformations. The traditional method LOF performs better than deep learning methods on CT and SAD.

4) *Ablation Study on DCL Loss Terms:* We investigate in an ablation study how the DCL manages the trade-off between semantics and diversity discussed in Section II-B1 (for a detailed discussion see also Appendix A.4, available online). This ablation also demonstrates empirically that when it comes to learning transformations for anomaly detection, the DCL is better suited than the contrastive loss or the classification loss. To study the trade-off between semantics and diversity we introduce a hyperparameter β to adjust the relative importance of the two terms in the DCL:

$$l_k(x) = -\log h(T_k(x), T_0(x)) + \beta \log \sum_{T_l \in \mathcal{T} \setminus \{T_k\}} h(T_k(x), T_l(x)).$$

In our default DCL ($\beta = 1$), the weight between the two objectives is balanced. To analyze the effect of this weighting, we perform an ablation study with $\beta = 0.1$ and $\beta = 10$. A lower β (e.g., 0.1) reduces the emphasis on the diversity objective, while a higher β (e.g., 10) reduces the emphasis on aligning with the original data.

To evaluate these objectives, we introduce the following metrics:

- **Semantics** $\frac{1}{K} \sum_{k=1}^K h(T_k(x), T_0(x))$:² This score measures how similar the transformed embeddings are to the embedding of the original data.
- **Diversity** $\frac{1}{K(K-1)} \sum_{k=1}^K \sum_{T_l \in \mathcal{T} \setminus \{T_k, T_0\}} \frac{1}{h(T_k(x), T_l(x))}$, which measures the dissimilarity between the transformed embeddings. A higher score indicates greater diversity among the transformations.
- **Semantics $\times$ Diversity**: a product of semantics and diversity scores evaluates the balance between the two objectives. A high product value indicates that both objectives are well balanced.

² $h(\cdot)$ is the exponential function of the cosine similarity

TABLE IV

AN ABLATION STUDY COMPARING THE CONTRASTIVE LOSS, THE CLASSIFICATION LOSS, AND THE DCL WITH VARIOUS WEIGHTS BETWEEN THE TWO LOG TERMS IN THE LOSS ON LEARNING TRANSFORMATIONS FOR ANOMALY DETECTION

	Semantics	Diversity	Semantics×Diversity	Anomaly detection AUC
Contrastive loss	2.72	0.37	1.00	0.892
Classification loss	1.10	0.83	0.92	0.915
DCL (default $\beta = 1$)	1.69	0.80	1.36	0.989
DCL ($\beta = 0.1$)	2.72	0.37	1.00	0.781
DCL ($\beta = 10$)	1.00	1.08	1.08	0.983

Notice that DCL ($\beta = 1$) is our default loss. We show that our default DCL ($\beta = 1$) learns transformations that best manage the trade-off of aligning with the original data and being diverse to other transformed views, and achieves the best anomaly detection performance. The experiment is run on the time series data (spoken arabic digits).

We fit this loss to the arabic digits in the SAD data. As shown in Table IV, our default DCL ($\beta = 1$) achieves the highest product of the semantics and diversity scores, indicating the best balance between these objectives. In contrast, the contrastive loss learns transformations (with a semantics score of $\exp(1) = 2.72$ and a diversity score of $\exp(-1) = 0.37$) that are perfectly aligned with the original data but sacrifices diversity, the classification loss learns transformations that are diverse but share much less relevant information with original data than others. Similarly, the DCL ($\beta = 0.1$) achieves a higher semantics score but sacrifices diversity, the DCL ($\beta = 10$) achieves a higher diversity score but compromises on semantics. This empirical result aligns with our theoretical analysis, confirming that the default DCL ($\beta = 1$) best balances the objectives of aligning with the original data while maintaining diversity among transformed views, ultimately resulting in superior anomaly detection performance.

C. Anomaly Detection for Image Data

Anomaly detection on images is important for many machine learning applications and has received a lot of attention in the last few years. Recent work [10], [63], [74] employs shallow anomaly detectors (KNN or OCSVM) upon the image features obtained from a pretrained model often achieving better detection accuracy than self-supervised anomaly detection on raw images. This raises the question of how to apply self-supervised anomaly detection to image features which might lead to an additional performance boost. However, applying self-supervised anomaly detection methods (such as Golan and El-Yaniv [9], [25], [82]) to image features is difficult since all of them rely on hand-crafted data augmentations designed for raw images. Hand-crafting transformations requires intuition about image invariances (e.g., humans still recognize dogs in rotated images of dogs). These intuitions do not readily generalize to image features. In contrast, NeuTraL AD learns transformations and hence requires no intuition, enabling a direct deployment on image features.

In this Section, we first present how to apply neural transformations to raw images and image features. We then show that when applying NeuTraL AD on image features, we learn meaningful and semantically diverse transformations and achieve competitive anomaly detection accuracy. We finally compare NeuTraL AD to strong baselines on large-scale out-of-distribution benchmarks.

1) *Neural Transformation Learning for Image Data:* NeuTraL AD learns data-dependent transformations automatically. Therefore, we can apply neural transformations to raw images or

features obtained from existing pretrained feature extractors. For both variants, we define the similarity function as the cosine similarity $\text{sim}(a, b) := a^T b / \|a\| \|b\|$, set the temperature $\tau = 0.1$. T_0 is modeled by a composition of an identity transformation and an encoder f . The transformation T_k with $k \in \{1, \dots, K\}$ is modeled by a composition of a transformation function ϕ_k and the shared encoder f .

Neural Transformation Learning on Raw Images (NeuTraL-I): We consider a dataset of images x . The transformation function is parametrized multiplicatively as $\phi_k(x) = x \odot M_k(x)$, where $M_k(x)$ is an attention mask (with values between 0 and 1) and the multiplication is applied element-wise. M_k is modeled by a residual convolutional neural network with affine-free batch normalization layers and a final sigmoid activation. All bias terms in the network are fixed as zero. We set the number of neural transformations $K = 15$ for all image datasets. As in Bergman and Hoshen [9], we use a ResNet architecture as the encoder f . Its output dimension is 64, and its parameters are learned together with the parameters of the transformation functions.

Neural Transformation Learning on Image Features (NeuTraL-F): We process images with a pretrained model and obtain a dataset of image features x as in Bergman et al. [10]. We consider two parametrizations of the transformation function: *forward* $\phi_k(x) := M_k(x)$ and *multiplicative* $\phi_k(x) := M_k(x) \odot x$. $M_k(x)$ is computed by three bias-free linear layers with affine-free batch normalization layers and ReLU activations. A sigmoid activation is applied on the final layer of M_k when using the multiplicative parametrization. We set the number of neural transformations $K = 15$. The encoder f is modeled by two linear layers with an intermediate ReLU activation. The output dimension of the encoder is 256.

2) *Anomaly Detection Experiments. Anomaly detection datasets:* We study three image datasets, F-MINST, CIFAR-10, and CIFAR-100, which are commonly used in prior works e.g., [10], [25], [63], [64], [74]. For CIFAR-100, we use 20 super-class labels in the experiments. On F-MNIST, we study the one-vs-rest setting. On the more challenging datasets, CIFAR-10 and CIFAR-100, we consider both the one-vs-rest setting and n -vs-rest setting with $n = \mathcal{N} - 1$.

Anomaly detection baselines: We compare NeuTraL AD with modern baselines based on different techniques. We include one deep anomaly detection baseline, deep SVDD [64], which learns a compact boundary in the embedding space to detect anomalies. As self-supervised baselines, we include Geometric-Transformation Classification (GEOM) [25] which designs an

TABLE V
AVERAGE AUCs (%) WITH STANDARD DEVIATIONS ON IMAGE DATASETS

Datasets	deep SVDD	GEOM	Contra-DA	DN2	CSI	PANDA	NeuTraL-I (ours)	NeuTraL-F (ours)
F-MNIST	92.8	93.5	94.8±0.3	94.4	-	95.6	94.5±0.1	94.9±0.04
CIFAR-10	64.8	86.0	92.5±0.6	92.5	94.3	96.2	69.6±0.3	95.3±0.06
CIFAR-100	-	78.7	86.5±0.7	89.3	89.6	94.1	68.1±0.1	93.2±0.12

We bold the best results. NeuTraL-F performs comparably to PANDA, which relies on the fine-tuning of the large pretrained feature extractor and a costly detection. Additional results with stronger feature extractors are in Appendix B, available online.

auxiliary transformation classification task for anomaly detection, and CSI which relies on contrastive learning. We also include three baselines, which build anomaly detectors upon image features. They are Distribution-Augmented Contrastive Learning (Contra-DA) [74], which proposed to use traditional anomaly detectors (e.g., OCSVM) on features from an encoder pretrained with a contrastive loss, Deep Nearest-Neighbors (DN2) [10] using a KNN based detector on features from a ResNet pretrained on ImageNet, and a state-of-the-art method PANDA [63], which improves over DN2 by finetuning the pretrained ResNet.

Empirical results: The results on image datasets with the one-vs-rest setting are reported in Table V, where the results of baselines are taken from the original papers. When neural transformations are applied to raw images, NeuTraL-I performs better than deep SVDD but worse than GEOM using hand-crafted transformations. It turns out that learning good transformations from the noisy raw images is still hard, and the learned transformations are not competitive with transformations selected by human experts. One reason for the poor performance on raw images could be the learned transformations affect mostly low-level features and do not interact with higher-level semantic features, as would be desired for a strong self-supervision task. This motivates applying NeuTraL AD to pretrained image features.

NeuTraL-F outperforms DN2 that uses the same pretrained feature extractor, and all baselines that have no access to a feature extractor pretrained on ImageNet. PANDA outperforms NeuTraL-F using the same pretrained feature extractor but also requires a careful finetuning of the large ResNet model and has a risk of performance degradation caused by catastrophic forgetting [63]. Additional stronger results of NeuTraL-F using the other pretrained feature extractor are provided in Appendix B, available online. Moreover, using PANDA at test-time is expensive; the KNN-based detection is slower than the forward-pass of the other methods in Table V and it requires storing the training data for test time. Our method achieves the highest performance among all methods that do not suffer from this drawback. When using PANDA with a more efficient distance for detection, the results will be 2% lower (e.g., CIFAR-10: from 96.2% to 94.2%) [63] and then outperformed by NeuTraL-F (95.3%). It is very hard even for human experts to design good transformations on the feature vectors, while NeuTraL AD can learn them from data automatically. NeuTraL AD offers a convenient way to do self-supervised learning on the image features to utilize the powerful existing pretrained models continuously being developed and improved by the community.

What do the transformed features look like?

For an interpretable visualization of the transformed features learned by NeuTraL-F, we invert them back to the image space

by seeking an image that best matches the source representation [46]. From the resulting images shown in the first row of Fig. 5, we can see that the transformations disrupt different local regions and textures of the image but preserve the global shape of the object. The learned local disruptions preserve global semantics and vary, thereby satisfying the diversity requirement.

Furthermore, we analyze the semantic information contained in the transformed features by checking the class prediction of a downstream classifier trained on the raw features and ground truth class labels. In the second row of Fig. 5 we can see that the predictions given transformed features (blue) perfectly match the predictions given the raw features (orange).³ This confirms the claim that the learned transformations manipulate the image without changing the global semantics as encoded by the class label.

How do the methods cope with an increased variability of inliers? To study this, we consider the n -vs-rest setting with $n = \mathcal{N} - 1$. As shown in Table VII, NeuTraL-F (with forward parametrization of the transformations) outperforms baselines on CIFAR-10 but performs worse than DN2 on CIFAR-100. This is consistent with the results on time series in Table III, where the KNN-based methods (DN2 in vision and LOF in sequences) have an advantage in anomaly detection when the normal distribution contains multiple modes.

3) Out-of-Distribution (OOD) Experiments. OOD datasets: We follow the experiment setup of Li et al. [40] and test the methods on CIFAR-10, CIFAR-100, and ImageNet-30. When assuming CIFAR-10 is in distribution, we consider CIFAR-100, SVHN, and LSUN as OOD datasets. For CIFAR-100, OOD samples are from CIFAR-10, SVHN, and LSUN. For ImageNet-30, OOD samples are from Stanford Dogs, Places365, Oxford Flowers, Oxford Pets, Food101, Caltech256, and Describable Textures Dataset (DTD).

OOD baselines: We compare NeuTraL AD with four recent strong OOD baselines. They are three contrastive learning-based methods including SupContrast [35], CSI [75], SSD+ [71], and MOOD [40] which uses representations from a masked image model pretrained and finetuned on ImageNet.

Empirical results: We evaluate NeuTraL-F for OOD detection following the setup in Li et al. [40] and report the results in Table VI. NeuTraL-F is equipped with the image features from the finetuned masked image modeling models in MOOD [40]. As shown in Table VI, NeuTraL-F improves over the SOTA baseline MOOD on both CIFAR-10 and CIFAR-100.

³The transformed features have not been used to train the classifier.

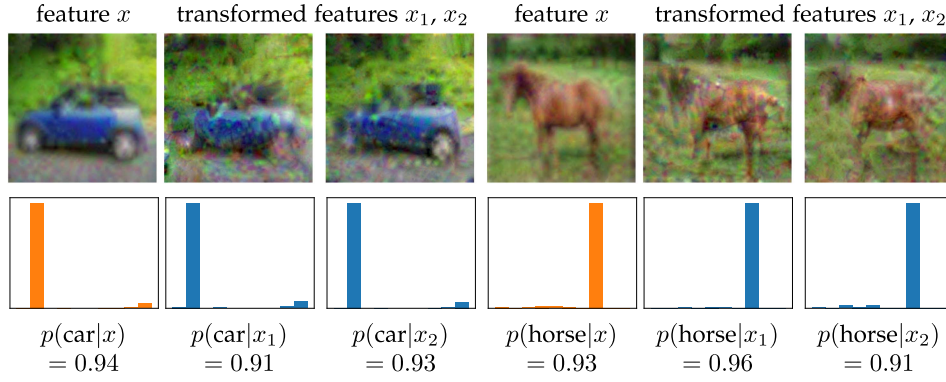


Fig. 5. Visualization and semantics checking examples from CIFAR-10. By inverting the original and transformed image features, we visualize the features in the image space in the first row. In the second row, we plot their class prediction results from a downstream classifier. The transformed features are diverse and still preserve the semantic information.

TABLE VI
AUCs (%) FOR MULTI-CLASS OOD DETECTION ON CIFAR-10, CIFAR-100, IMAGENET-30

In-Distribution Out-of-Distribution	CIFAR-10				CIFAR-100			
	SVHN	CIFAR-100	LSUN	AVG	SVHN	CIFAR-10	LSUN	Avg.
SupContrast	97.3	88.6	92.8	92.9	95.6	78.3	-	87.0
CSI	97.9	92.2	97.7	95.9	-	-	-	-
SSD+	99.9	93.4	98.4	97.2	98.2	78.3	79.8	85.4
MOOD	99.8	99.4	99.9	99.7	96.5	98.3	96.3	97.0
NeuTraL-F (ours)	100.0	100.0	99.77	99.92	99.99	100.0	99.99	99.99

(a) CIFAR

In-Distribution Out-of-Distribution	ImageNet-30							
	Dogs	Places365	Flowers102	Pets	Food	Caltech256	DTD	Avg.
SupContrast	95.6	89.7	92.2	94.2	81.2	90.2	92.1	90.7
CSI	98.3	94.0	96.2	97.4	87.0	93.2	97.4	94.8
MOOD	99.4	98.9	100.0	99.1	96.6	99.5	98.9	98.9
NeuTraL-F (ours)	99.6	98.8	100.0	99.5	96.9	99.7	97.1	98.8

(b) ImageNet

NeuTraL-F achieves the best results on CIFAR-10 and CIFAR100, and performs comparably to the SOTA baseline MOOD [40] on ImageNet-30.

TABLE VII
AVERAGE AUCs (%) WITH STANDARD DEVIATIONS ON CIFAR-10 AND CIFAR-100 UNDER n -VS-REST SETTING WITH $n = \mathcal{N} - 1$

Datasets	GEOM	DN2	NeuTraL-F (ours)
CIFAR-10	61.7	71.7	77.7$\pm$0.4
CIFAR-100	57.3	71.0	68.0 $\pm$ 0.2

NeuTraL AD outperforms the baselines on average.

NeuTraL-F also achieves decent and comparable results on the large-scale ImageNet-30 benchmark.

D. Anomaly Detection on Tabular Data

Tabular data is another important application area of anomaly detection. For example, many types of health data come in tabular form. To unleash the power of self-supervised anomaly detection for these domains, Bergman and Hoshen [9] suggest using random affine transformation. Here we study the benefit of *learning* the transformations on tabular data with NeuTraL AD. In this Section, we first present the datasets and baselines we use. We then present specific architecture choices and implementation details and finally describe the empirical results. We find that NeuTraL AD raises the detection accuracy by 2.9% on average in terms of F1-score.

1) *Datasets and Baselines: Tabular Datasets.* We base our empirical study of tabular anomaly detection on previous

work [9], [93] and follow their choice of datasets as well as their precedent of reporting results in terms of F1-scores. The datasets include the small-scale medical datasets Arrhythmia and Thyroid as well as the large-scale cyber intrusion detection datasets KDD and KDDRev. We follow the configuration of Bergman and Hoshen [9] to train all models on half of the normal data and test on the rest of the normal data as well as the anomalies.

- Arrhythmia: A cardiology dataset from the UCI repository contains 274 continuous and 5 categorical attributes.
- Thyroid: A medical dataset from the UCI repository contains attributes related to hyperthyroid diagnosis.
- KDD: KDDCUP99 10 percent dataset from the UCI repository contains 34 continuous and 7 categorical attributes.
- KDDRev: It is derived from the KDDCUP99 10 percent dataset. The non-attack samples are considered normal, and attack samples are considered abnormal.

On these 4 datasets we use the same preprocessing steps as Bergman and Hoshen [9].

Tabular Baselines: We compare NeuTraL AD to shallow anomaly detection baselines, including OCSVM [48], IF [43], and LOF [11], and to the deep anomaly detection methods deep SVDD [64], DAGMM [93], GOAD [9], and Deep Robust One-Class Classification (DROCC) [26]. The results of OCSVM,

TABLE VIII
F1-SCORES (%) WITH STANDARD DEVIATION FOR ANOMALY DETECTION ON
TABULAR DATASETS (CHOICE OF F1-SCORE CONSISTENT WITH PRIOR WORK)

	Arrhythmia	Thyroid	KDD	KDDRev	Avg.
OCSVM	45.8	38.9	79.5	83.2	61.9
IF	57.4	46.9	90.7	90.6	71.4
LOF	50.0	52.7	83.8	81.6	67.0
deep SVDD	53.9±3.1	70.8±1.8	99.0±0.1	98.6±0.2	80.6
DAGMM	49.8	47.8	93.7	93.8	71.3
GOAD	52.0±2.3	74.5±1.1	98.4±0.2	98.9±0.3	81.0
DROCC	47.5±0.7	75.6±1.8	92.9±0.4	95.2±1.0	77.8
NeuTraL AD (ours)	60.3±1.1	76.8±1.9	99.3±0.1	99.1±0.1	83.9

NeuTraL AD achieves the state-of-the-art.

LOF, DAGMM, and GOAD are from Bergman and Hoshen [9]. We obtained the results for deep SVDD and DROCC using the code with the respective publications.

2) *Neural Transformation Learning on Tabular Data*: We consider a dataset of tabular data $x \in \mathbb{R}^d$. T_0 is modeled by a composition of an identity transformation and an encoder f . The transformations T_k with $k \in \{1, \dots, K\}$ are modeled by a composition of a transformation function ϕ_k and the shared encoder f . We consider two parametrizations of the transformation function ϕ_k : *residual* $\phi_k(x) := M_k(x) + x$ with $M_k(x) \in \mathbb{R}^d$, and *multiplicative* $\phi_k(x) := M_k(x) \odot x$ with $M_k(x) \in (0, 1)^d$. M_k and the encoder f are both modeled by feed-forward neural networks. In the embedding space, we define the similarity function as the cosine similarity $\text{sim}(a, b) := a^T b / \|a\| \|b\|$.

Implementation Details: The neural network of M_k consists of two bias-free linear layers with an intermediate ReLU activation. When using the multiplicative parametrization, it has a sigmoid activation on the final layer. We use the residual parametrization for the neural transformations on Thyroid and Arrhythmia, and use the multiplicative parametrization on KDDCUP, and KDDCUP-Rev. The neural network of the encoder f consists of five bias-free linear layers with ReLU activations. The output dimensions of the encoder are 24 for Thyroid and 32 for the other datasets. We set the number of neural transformations $K = 11$ and the temperature $\tau = 0.1$ on all tabular datasets.

3) *Empirical Results*: In line with prior work, we use the averaged F1-score (%) with standard deviation as the evaluation metric and report the results in Table VIII. The results of OCSVM, LOF, DAGMM, and GOAD are taken from [9]. For deep SVDD and DROCC, we run their official implementations and report the results. NeuTraL AD outperforms all baselines on all tabular datasets. In particular, NeuTraL AD raises the F1-score on Arrhythmia by 3.7% and on Thyroid by 2.4%. Compared with the self-supervised baseline GOAD with random affine transformations, the neural transformations learned from data lead to better detection accuracy. In addition, we find that NeuTraL AD needs fewer transformations than GOAD. On the medical datasets, for example, GOAD uses 256 transformations, while NeuTraL AD achieves superior performance with only 11 transformations.

E. Anomaly Detection on Text

There are many beneficial applications of anomaly detection on text such as detecting spam emails, fake tweets, or other

anomalous content on the web. Here we present NeuTraL AD for text and compare it with recent approaches for sentence-level anomaly detection.

NeuTraL AD on text uses a pre-trained language model to preprocess the text. To create different views of each sentence, neural transformations are then applied to the list of word embeddings comprising each sentence. Below, we describe how the individual word embeddings are transformed and how an attention mechanism is used to aggregate word embeddings to create the different views of each sentence needed for the NeuTraL AD loss.

1) *Datasets and Baselines. Text Datasets*: We study NeuTraL AD on the *Reuters-21578* dataset, which is commonly used in previous text anomaly detection work [34], [48], [66]. We used the same pre-processing as in Ruff et al. [66], including lowercasing, removing stopwords, and tokenization. As in Ruff et al. [66], we select seven classes that have exactly one label. They are *earn*, *acq*, *crude*, *trade*, *money-fx*, *interest*, and *ship*. We evaluate NeuTraL AD on them with the one-vs-rest setting.

Text Baselines: The text anomaly detection baselines include deep SVDD [64], which aim to encompass normal data within a hypersphere in the embedding space, and Multi-Modal Deep Support Vector Data Description (mSVDD) [34] which extends the idea of deep SVDD to multiple hyperspheres. Further, the baselines include Deep Multi-Sphere Support Vector Data Description (DMSVDD) [23], a hard version of mSVDD, and Context Vector Data Description (CVDD) [66], which augments deep SVDD for text with multiple attention heads to reflect that there are multiple normal semantic contexts for each word. All baselines are built upon the same word embeddings. Specifically, all embeddings are extracted using GloVe 6B [57].

2) *Neural Transformation Learning on Text*: NeuTraL AD on text uses a pretrained language model to preprocess the text. To create different views of each sentence, neural transformations are then applied to the list of word embeddings comprising each sentence. Below, we describe how the individual word embeddings are transformed and how an attention mechanism is used afterwards to aggregate the word embeddings to create the different views of each sentence needed for its loss.

As in previous work [23], [34], [66], we preprocess the data with a language model such that each sentence x is represented by a sequence of word embeddings $x = [e_1, \dots, e_l] \in \mathbb{R}^{d \times l}$. The length l of the sentences is variable from sentence to sentence. T_0 is modeled by a composition of an identity transformation and an encoder f . The transformation T_k with $k \in \{1, \dots, K\}$ is modeled by a composition of a transformation function ϕ_k and the shared encoder f . The transformation function ϕ_k is applied to the individual word embeddings, producing the transformed embeddings,

$$x_k = \phi_k(x) = M_k(x) \odot x,$$

where M_k is a feed-forward neural network with a final sigmoid activation. It is applied to each embedding in the sentence separately, i.e., $M_k(x) = [M_k(e_1), \dots, M_k(e_l)] \in (0, 1)^{d \times l}$. After element-wise multiplication with the word embeddings, we have the transformed embedding $x_k = [M_k(e_1) \odot e_1, \dots, M_k(e_l) \odot e_l] \in \mathbb{R}^{d \times l}$. The embeddings are aggregated using attention

TABLE IX
AVERAGE AUCs (%) WITH STANDARD DEVIATIONS ON REUTERS DATASETS

Class	DSVDD	CVDD	DMSVDD	mSVDD	NeuTraL AD (ours)
earn	95.9	91.8	96.0	95.9	97.4±0.1
acq	89.4	91.5	89.8	90.1	91.6±1.1
crude	92.8	95.5	92.1	92.4	96.4±0.4
trade	98.4	99.2	98.8	98.6	99.5±0.1
money-fx	86.3	82.8	87.1	87.1	89.8±0.1
interest	97.3	97.7	97.2	97.3	98.6±0.4
ship	92.5	95.6	93.0	92.6	89.6±0.7
average	93.2	93.1	93.4	93.4	94.7±0.1

NeuTraL AD achieves the best average AUC and the best accuracy on 5 of 7 experimental variants.

TABLE X
EXAMPLE OF WORDS WITH LARGE ATTENTION WEIGHTS FOR EACH CLASS

earn	acq	crude	trade	money-fx	interest	ship
<i>pretax</i>	<i>share</i>	<i>futures</i>	<i>imports</i>	<i>dollar</i>	<i>rate</i>	<i>ship</i>
<i>rupees</i>	<i>dollars</i>	<i>crude</i>	<i>exports</i>	<i>currency</i>	<i>pct</i>	<i>ships</i>
<i>dlrs</i>	<i>billion</i>	<i>oil</i>	<i>trade</i>	<i>bank</i>	<i>rates</i>	<i>port</i>

The attention mechanism of NeuTraL AD learns to highlight words with a similar semantic content.

$A(x) \in (0, 1)^l$ and are then encoded. In summary,

$$f(x_k) = g(x_k A(x))$$

$$\text{with } A(x) = \text{sigmoid}(\tanh(x^T W_1) W_2), \quad (8)$$

where W_1 and W_2 are learnable weights, and g is a feed-forward neural network that maps the aggregated embeddings to a low dimensional embedding space.⁴ In this embedding space, we define the similarity function as the cosine similarity $\text{sim}(a, b) := a^T b / \|a\| \|b\|$.

Implementation Details: For M_k we use a neural network of three bias-free linear layers of units [300,300,300] with ReLU activations and affine-free batch normalization layers and a sigmoid activation on the top. W_1 and W_2 have the shapes of [300,300] and [300,1]. For g we use a neural network of three bias-free linear layers of units [300,100,100] with ReLU activations. We set the number of transformations $K = 10$ and the temperature $\tau = 0.1$.

3) Empirical Results: The anomaly detection results in terms of average AUC (%) with standard deviation are reported in Table IX. NeuTraL AD achieves the best detection accuracy on 6 of 7 experimental variants. On average, NeuTraL AD outperforms all baselines by at least 1.3% in terms of AUC. In Table X, we record the three most relevant words (in terms of highest attention weight $A(x)$) for each class. We can see that the attention mechanism of NeuTraL AD learns to highlight words with similar semantic content.

F. Graph-Level Anomaly Detection

Graphs offer a powerful representation for many types of structured data, including chemical processes, molecules, financial or social networks. There is much work focused on the task of detecting anomalous nodes and edges within a graph [5]. However, in many applications, it is much more relevant to ask whether an entire graph is abnormal.

In a financial network, for example, with nodes representing individuals, businesses, and banks and the edges representing transactions, it might be difficult to detect all criminal activity by looking at individual nodes and edges. By using tools for graph-level anomaly detection, we might be able to detect an entire criminal network rather than suspicious individual entities. In alternative healthcare applications, one might be interested in detecting novel (anomalous) molecules whose atoms and bonds are nodes and edges in the molecular graphs. Thus, while every atom may be known, the molecule might be novel as a whole. This, and many other applications, can profit from automated methods for detecting abnormal graphs.

For applying self-supervised methods to graph-level anomaly detection, good graph transformations are required. You et al. [87] observe that different graph datasets require different transformation types. For example, edge perturbation benefits representation learning on social networks but hurts the quality of representations of molecules learned from biochemical graph data. Here NeuTraL AD has a conceptual advantage; the appropriate transformations are directly learned from the data.

We demonstrate that this advantage makes neural transformation learning a useful tool for graph-level anomaly detection, significantly outperforming existing methods. We use it to analyze graphs from various important domains, including bioinformatics, molecular data, and social networks.

1) Datasets and Baselines. Graph Datasets: We study NeuTraL AD on six graph classification datasets that are representative of three domains. They are two bioinformatics datasets: DD and PROTEINS, two molecular datasets: NCI1 and AIDS, and two datasets of social networks: IMDB-BINARY and REDDIT-BINARY. In IMDB-BINARY, node attributes are derived by encoding the degree of each node in a one-hot vector. In REDDIT-BINARY, node attributes are set to a constant. All datasets are made available by Morris et al. [51].

Following the one-vs-rest evaluation setup, we create $\mathcal{N}$ experimental variants for each dataset ($\mathcal{N}$ is the number of classes). We use 10-fold cross-validation to estimate the model performance. In each fold, 10% of the training set is held-out for validation. Results are reported in terms of average AUC (%) with standard deviation averaged over 10 folds and over all experimental variants.

In addition, all methods are evaluated in terms of their susceptibility to performance flip. Zhao and Akoglu [90] coined the term ‘‘performance flip’’ for anomaly detection benchmarks derived from binary classification datasets. An effective graph-level anomaly detection method should not suffer from the performance flip issue.

Definition 1. (Performance flip [90].) A model suffers from performance flip on an anomaly detection benchmark derived from a classification dataset if it performs worse than random on at least one experimental variant.

Graph Baselines: To build a comprehensive benchmark for graph-level anomaly detection, we include both graph Neural Network (GNN)-based methods and non-GNN-based methods as baselines. Two GNN-based baselines are One-Class Graph Isomorphism Network (OCGIN) [90], extending the idea of deep

⁴For the untransformed x , (8) becomes $f(x) = g(xA(x))$.

TABLE XI
AVERAGE AUCs (%) WITH STANDARD DEVIATIONS FOR GRAPH-LEVEL ANOMALY DETECTION ON SIX DATASETS FROM THREE DOMAINS

Datasets	DD	PROT	NCI1	AIDS	IMDB-B	RDT-B	Avg.
WLK	50.2±0.3*	49.7±0.5*	49.6±0.4*	51.3±0.8*	62.0±2.0	50.2±0.2*	52.2
PK	51.2±2.3*	50.8±1.5*	51.4±1.7*	59.5±2.3*	53.5±2.0	50.0	52.7
Graph2Vec	49.5±2.2*	53.2±3.0*	50.5±0.7*	48.4±0.8*	54.3±1.6	51.5±0.6*	51.2
FGSD	66.0±2.3	58.5±1.8*	55.4±0.7	91.6±3.7	57.1±1.8	-	65.7
GTP	54.2±1.9	61.9±2.9	55.3±1.2*	77.2±2.9	57.6±1.1	64.4±2.2	61.8
OCGIN	50.7±1.2*	54.2±1.2*	53.6±1.3*	60.8±2.2*	60.4±2.8	67.1±3.5	57.8
NeuTraL AD (ours)	70.2±2.7	64.4±4.9	65.7±1.3	97.4±1.9	63.1±1.8	78.5±2.1	73.2

Results marked * perform worse than random on at least one experimental variant (performance flip). NeuTraL AD outperforms all baselines on all datasets and has no performance flip.

SVDD to graph-level anomaly detection, and Graph Transformation Prediction (GTP), a self-supervised detection baseline proposed by us based on graph transformation prediction (see details in Appendix B, available online).

Four non-GNN-based methods are two-stage detection methods using different feature extractors and a downstream OCSVM [48] for anomaly detection. Two of them use unsupervised graph embedding methods, Graph-to-Vector (Graph2Vec) [52] or Family of Graph Spectral Distances (FGSD) [79], to extract graph-level representations. The other two make use of graph kernels (Weisfeiler-Leman Subtree Kernel (WLK) [73] or Propagation Kernel (PK) [53]), which measure the similarity between graphs. Details are in Appendix B, available online.

2) *Neural Transformation Learning on Graphs*: We consider a dataset of attributed graphs x . For graph data, the neural transformations T_k are parameterized by GNNs. Each neural transformation T_k is an individual GNN, which transforms the graph and maps it to the embedding space.

Since graphs have varying sizes, the number of nodes can be an informative feature for distinguishing graphs and detecting abnormal ones. By using an appropriate GNN architecture for the neural transformations, we can introduce a correlation between the norm of embeddings and graph size. To ensure that this information is not lost, we use the negative euclidean distance as the similarity function $\text{sim}(a, b) := -\|a - b\|_2$ for the DCL.⁵ The resulting multi-view deep OCC loss (7) is a special case of the DCL. In Section II-C, we discuss how it is related to deep OCC. The results reported support the arguments that our method is more informative and more flexible than deep OCC.

Implementation Details: We use Graph Isomorphism Networks (GINs) [85] as T_k to extract graph-level representations. The network has 4 GIN layers, each of which includes two bias-free linear layers with an intermediate ReLU activation. On datasets from molecular and social-networks domains, graph normalization [12] is applied after each GIN layer. A readout function modeled by two bias-free linear layers with an intermediate ReLU activation and an add pooling layer is applied on each layer to get the graph-level representation on each hierarchical level. As in Xu et al. [86], the final graph-level representation of dimension 128 is obtained by concatenating all layer-wise graph-level representations. None of the transformations use biases, except the network T_0 , which adds a trainable bias term to the final graph-level representations. On all graph datasets, we set the number of transformations $K = 5$.

⁵The cosine similarity only considers angles between embeddings, thereby ignoring the information encoded in their norm.

3) *Empirical Results*: The anomaly detection results in terms of average AUC (%) with standard deviation are reported in Table XI. We can see that NeuTraL AD outperforms all baselines on all datasets and raises the detection accuracy in terms of AUC by 7.5% on average. NeuTraL AD improves over the previous deep method OCGIN, which is an extension of deep SVDD on graph-level anomaly detection, by 15.4% on average. We can empirically conclude that NeuTraL AD is more sensitive to anomalies than OCGIN since it extracts features from different views. The results with performance flip are marked with a * (it is preferable for a method not to suffer from performance flip). We can see that all baselines suffer from the performance flip issue, while NeuTraL AD is the only model without performance flip on any of the datasets. We provide the detailed results on both experimental variants of bioinformatic and molecular datasets in Appendix C, available online.

IV. RELATED WORK

Here we relate NeuTraL AD to existing work in deep anomaly detection, self-supervised learning, and the more specific contrastive representation learning.

Deep Anomaly Detection: Recently, there has been a rapidly growing interest in developing deep-learning approaches for anomaly detection [67]. While deep learning—by removing the burden of manual feature engineering for complex problems—has brought about tremendous technological advances, its application to anomaly detection is rather recent. Related work on deep anomaly detection includes deep autoencoder variants [16], [59], [91], deep OCC [22], [65], [66], multi-sphere OCC [23], [34], [66], [84], deep generative models [18], [69], and outlier exposure [30], [39], [61].

Self-supervised anomaly detection has led to drastic improvements in detection accuracy [25], [31], [74], [75], [82]. For images, Golan and El-Yaniv [25] and Wang et al. [82] augment images and learn to predict which transformation has been applied. After training, the resulting classifier is used for anomaly detection. An alternative approach is to train a model with a contrastive loss to tell if two views are of the same original image. This leads to strong representations [15], which can be used for anomaly detection [74], [75]. For time series anomaly detection, Zhou et al. [36], [81], [92] combine autoencoder-based methods or deep OCC with contrastive learning based on hand-crafted data augmentations. Following the work of Qiu et al. [60], several recent studies [13], [14], [54], [70] integrate contrastive learning with neural transformations

on times series. In the domain of tabular anomaly detection, Bergman and Hoshen [9] design an auxiliary classification tasks using random affine transformations, while ICL [72], a contemporaneous work of NeuTraL AD, proposes contrasting two complementary subsets of the tabular data. For graph-level anomaly detection, Luo et al. [45] employ contrastive learning with two perturbed graph encoders, while Ai et al. [3] design two types of topology augmentations for graph-level contrastive learning. Unlike all prior works, NeuTraL AD learns transformations that are effective for self-supervised anomaly detection on various data types.

Self-Supervised Learning: Self-supervised learning typically relies on data augmentation for auxiliary tasks [20], [24], [49], [55], [88]. The networks trained on these auxiliary tasks (e.g., patch prediction [20], solving jigsaw-puzzles [55], cross-channel prediction [88], or rotation prediction [24]) are used as feature extractors for downstream tasks. While many of these methods are developed for images, Misra et al. [49] propose temporal order verification as an auxiliary task for self-supervised learning of time series representations.

Contrastive Representation Learning: Many recent self-supervised methods have relied on the InfoMax principle [32], [42]. These methods are trained on the task to maximize the Mutual Information (MI) between the data and their context [56] or between different “views” of the data [6]. Computing the mutual information in these settings is often intractable, and various approximation schemes and bounds have been introduced [78]. By using noise contrastive estimation [27], [28] to bound MI, Oord et al. [56] bridge the gap between contrastive losses for MI-based representation learning and the use of contrastive losses in discriminative methods for representation learning [6], [15], [21], [29], [50]. We also use a contrastive loss. But while the contrastive loss of Chen et al. [15] (which is used for anomaly detection of images in Sohn et al. [74], [75],) contrast two views of the same sample with views of other samples in the minibatch, NeuTraL AD is tasked with determining the original version of a sample from different views of the same sample. The dependence on only a single sample is advantageous for scoring anomalies at test time and enables us to learn the data transformations.

Learning Data Augmentation Schemes: The idea of learning data augmentation schemes is not new. “AutoAugmentation” has usually relied on composing hand-crafted data augmentations [17], [33], [41], [62], [89]. Tran et al. [77] learn Bayesian augmentation schemes for neural networks, and Wong and Kolter [83] learn perturbation sets for adversarial robustness. Though their setting and approach are different, our work is most closely related to Tamkin et al. [76], who study how to generate views for representation learning in the framework of Chen et al. [15]. They parametrize their “viewmakers” as residual perturbations, which are trained adversarially to avoid trivial solutions where the views share no semantic information with the original sample.

NeuTraL AD falls into the area of deep, self-supervised anomaly detection, with the core novelty of learning the transformations so that we can effectively use them for anomaly detection on general data types.

V. CONCLUSION

We propose NeuTraL AD, a self-supervised anomaly detection method with learnable transformations. The key ingredient is a novel training objective based on a DCL, which encourages the learned transformations to be diverse and semantically meaningful. This unleashes the power of self-supervised anomaly detection to various data types. We show that NeuTraL AD generalizes deep OCC, leading to a more powerful and more flexible anomaly detection method thanks to the learnable transformations. Our extensive empirical study finds that NeuTraL AD as a unified anomaly detection method improves over many strong baselines on various non-image data types, including time series, tabular data, text data, and graphs, and achieves competitive results on images.

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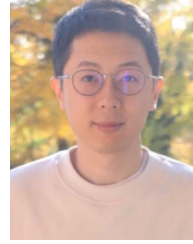
The Bosch Group is carbon neutral. Administration, manufacturing and research activities do no longer leave a carbon footprint. This also includes GPU clusters on which the experiments have been performed.

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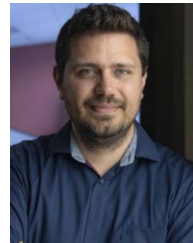
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